Riccardo **Orlando**

Ph.D. in Deep Learning and Natural Language Processing

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Experience

Researcher
Rome, Italy

Sapienza University of Rome — *Nov. 2021 - Jan. 2025*

- Led model pretraining and data preprocessing for Minerva (1B, 3B, and 7B) LLMs, the first family of Open Foundation Models trained from scratch on Italian data.
- Developed ReLiK, a novel retrieval-augmented system achieving SOTA performance in Entity Linking and Relation Extraction.
- Led the data annotation process that resulted in a novel dataset for Semantic Role Labeling.
- Developed production-ready pipelines for several state-of-the-art NLU tasks.
- Served as Teaching Assistant for MSc courses in Natural Language Processing.

NLP Engineer
Rome, Italy

Babelscape — *Feb. 2021 - Nov. 2021*

- Designed and implemented end-to-end multilingual NLP pipelines for production deployment.
- Built scalable ML serving framework that drastically reduced model deployment time.
- Authored and presented two papers at EMNLP 2021.

Software Engineer
Rome, Italy

Capgemini — *Mar. 2018 - Sept. 2018*

- Developed a back-end application using Spring Boot to handle internal workflows for different customers.
- Contributed to front-end development using Angular.

Education

Ph.D. in Computer Science (Deep Learning & Natural Language Processing)
Rome, Italy

Sapienza University of Rome — *Nov. 2021 - Jan. 2025*

- Thesis: Enhancing Semantic Understanding Across Multiple Dimensions: Towards A Unified Framework for Semantic Knowledge Extraction.
- Advisor: Roberto Navigli

Master of Science in Engineering in Computer Science
Rome, Italy

Sapienza University of Rome — *Oct. 2018 - Jan. 2021*

- Final Grade: 110/110 with honors
- Thesis: Automatic Approaches to Produce Multilingual Training Data for Semantic Role Labelling.

Bachelor of Science in Computer Science
Rome, Italy

Sapienza University of Rome — *Oct. 2014 - May 2018*

- Final Grade: 101/110
- Thesis: Densest Subgraph in Fork/Join.

Relevant Publications

2024	Francesco Maria Molfese, Simone Conia, Riccardo Orlando, and Roberto Navigli. **ZEBRA: Zero-Shot Example-Based Retrieval Augmentation for Commonsense Question Answering**.	*EMNLP 2024*
2024	Riccardo Orlando, Luca Moroni, Pere-Lluís Huguet Cabot, Edoardo Barba, Simone Conia, Sergio Orlandini, Giuseppe Fiameni, and Roberto Navigli. **Minerva LLMs: The First Family of Large Language Models Trained from Scratch on Italian Data**.	*CLiC-it 2024*
2024	Riccardo Orlando, Pere-Lluís Huguet Cabot, Edoardo Barba, and Roberto Navigli. **ReLiK: Retrieve and LinK, Fast and Accurate Entity Linking and Relation Extraction on an Academic Budget**.	*ACL 2024*
2024	Simone Conia, Edoardo Barba, Abelardo Carlos Martínez Lorenzo, Pere-Lluis Huguet Cabot, Ricardo Orlando, Luigi Procopio, Roberto Navigli. **MOSAICo: a Multilingual Open-text Semantically Annotated Interlinked Corpus**.	*NAACL 2024*
2023	Riccardo Orlando, Simone Conia, and Roberto Navigli. **Exploring Non-Verbal Predicates in Semantic Role Labeling: Challenges and Opportunities**.	*ACL 2023*
2022	Riccardo Orlando, Simone Conia, Stefano Faralli, and Roberto Navigli. **Universal Semantic Annotator: the First Unified API for WSD, SRL and Semantic Parsing**.	*LREC 2022*

2021	Riccardo Orlando, Simone Conia, Fabrizio Brignone, Francesco Cecconi, and Roberto Navigli. **AMuSE-WSD: An All-in-one Multilingual System for Easy Word Sense Disambiguation**.	*EMNLP 2021*
2021	Simone Conia, Riccardo Orlando, Fabrizio Brignone, Francesco Cecconi, and Roberto Navigli. **InVeRo-XL: Making Cross-Lingual Semantic Role Labeling Accessible with Intelligible Verbs and Roles**.	*EMNLP 2021*

Skills

Frameworks Deep Learning & Machine Learning (PyTorch, PyTorch Lightning, Hydra, Transformers, ONNX, TensorFlow, scikit-learn)
Back-end FastAPI, Django, REST API
Languages Python, Java, Kotlin, Bash/Shell
Developer Tools Git, Docker, VS Code, Visual Studio, PyCharm, IntelliJ

Honors & Awards

2024	**Winner**, 3x Italian SuperComputing Resource Allocation (ISCRA) Call for Proposal type C for compute hours at CINECA, the Italian HPC Center	*Rome, Italy*
2023	**Winner**, Italian SuperComputing Resource Allocation (ISCRA) Call for Proposal type B for compute hours at CINECA, the Italian HPC Center	*Rome, Italy*
2022	**Winner**, Call for Proposals for Scientific Research	*Rome, Italy*

Program Committees

2024 **Reviewer**, ARR February 2024, ARR June 2024
2023 **Reviewer**, ARR December 2023, ARR October 2023, EMNLP 2023, ACL 2023, EACL 2023
2022 **Reviewer**, LREC 2022
2021 **Reviewer**, EMNLP 2021